



Parktown Boys' High School

Geography Department

Grade 12 June Exam

Examiner: Mr A Meintjes
Moderator: Mr D Grace

Time: 180 Minutes
Total: 150 Marks
Date: 19 June 2023

INSTRUCTIONS AND INFORMATION

1. This question paper consists of TWO SECTIONS.

SECTION A

QUESTION 1: CLIMATE AND WEATHER (40 MARKS)

QUESTION 2: GEOMORPHOLOGY (40 MARKS)

QUESTION 3 : SETTLEMENT GEOGRAPHY (40 MARKS)

SECTION B

QUESTION 4: GEOGRAPHICAL SKILLS AND TECHNIQUES (30 MARKS)

Answer ALL FOUR questions.

2. The question paper consists of FOUR questions.
3. ALL diagrams are included in the question paper.
4. Where possible, illustrate your answers with labelled diagrams.
5. Leave a line between subsections answered.
6. Start EACH section at the top of a NEW page.

7. Number your answers correctly according to the numbering system used in this question paper
8. Do NOT write in the margins of your ANSWER BOOK.
9. Write neatly and legibly.
10. In SECTION B you are provided with a 1 : 50 000 topographic map (2931 CA VERULAM) and an orthophoto map (2931 CA 11 VERULAM) of a part of the mapped area.
11. Show ALL calculations and use the formulae provided, where applicable. Marks will be allocated for these.
12. Indicate the unit of measurement in the final answer of calculations, e.g. 10 km; 2,1 cm.
13. You may use a non-programmable and a magnifying glass.
14. The area demarcated in RED and BLACK on the topographic map represents the area covered by the orthophoto map.

SECTION A: CLIMATE AND WEATHER, GEOMORPHOLOGY, AND SETTLEMENTS

QUESTION 1: Climate and Weather

1.1. Various possible answers are provided for each question. Write down **only the letter** of the correct answer next to the corresponding number.

1.1.1. Air that flows down a mountain and along the valley floors at night is known as a(an) ...

- A. Upslope wind.
- B. Anabatic wind.
- C. Downslope wind.
- D. Katabatic wind. (1)

1.1.2. Which direction should one face their house if they wanted to benefit from more heat in the South African winter?

- A. North
- B. West
- C. South
- D. East (1)

1.1.3. A cold front occlusion happens when....

- A. The cold air is left on the ground.
- B. Air ahead of the cold front is slightly warmer than the air behind the cold front.
- C. Air ahead of the cold front is colder than the air behind the cold front.
- D. The air ahead of the warm front is incredibly hot. (1)

1.1.4. Which of the following is NOT associated with the formation of a tropical cyclone?

- A. Coriolis force
- B. Mixing of the westerlies and polar easterlies
- C. Latent heat
- D. Hot moist air (1)

1.1.5. Which of the following factors would hinder the development of an inversion layer?

- A. Low temperatures
- B. Clear skies
- C. Cloudy skies
- D. Still conditions

(1)

1.2. Match the definition/description in **Column B** with each term in **Column A**.

Write the **letter only** of the correct answer next to the corresponding number.

E.g. 2.9. M

Column A	Column B
1.2.1. Moisture front	A. An area on earth where no tropical cyclones form.
1.2.2. Equator	B. The average atmospheric conditions of a small area such as a valley.
1.2.3. Isobars	C. Warm, moist north-easterly winds meet cool south-westerly winds here
1.2.4. Inversion layer	D. A layer of air where the temperature increases with height.
1.2.5. Microclimate	E. Lines that measure the average temperature of an area
	F. An area that does not receive any sunlight
	G. Lines measuring the hectopascals or millibars of an area
	H. The average atmospheric conditions of a large area such as Johannesburg.

(5 x 1) (5)

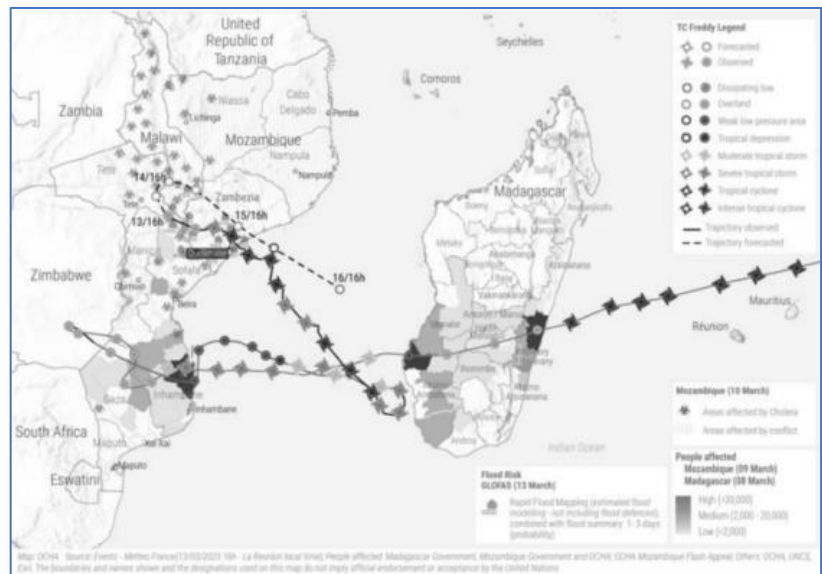
1.3 Refer to the article below on Tropical Cyclone Freddy.

Record-breaking Tropical Cyclone Freddy formed on Feb. 3, 2023, northwest of Australia before traveling across the entire Indian Ocean to become the first Category 5 storm of this Southwest Indian Ocean cyclone season.

According to the National Oceanic and Atmospheric Administration, no tropical cyclones have taken such a path across the Indian Ocean in the past two decades. On March 7, Freddy became the longest-lived tropical cyclone ever recorded and is now officially Earth's most energetic storm ever observed.

Cyclone Freddy made landfall on Madagascar's eastern coast, near Mananjary, on Feb. 21. Freddy then moved across the Mozambique Channel and made landfall in Mozambique's Inhambane province on Feb. 24.

Cyclone Freddy made its third landfall in total and its second landfall in Mozambique on March 11 in Zambézia province, with maximum winds of almost 92 miles per hour (148 kilometers per hour). Freddy moved over land as a

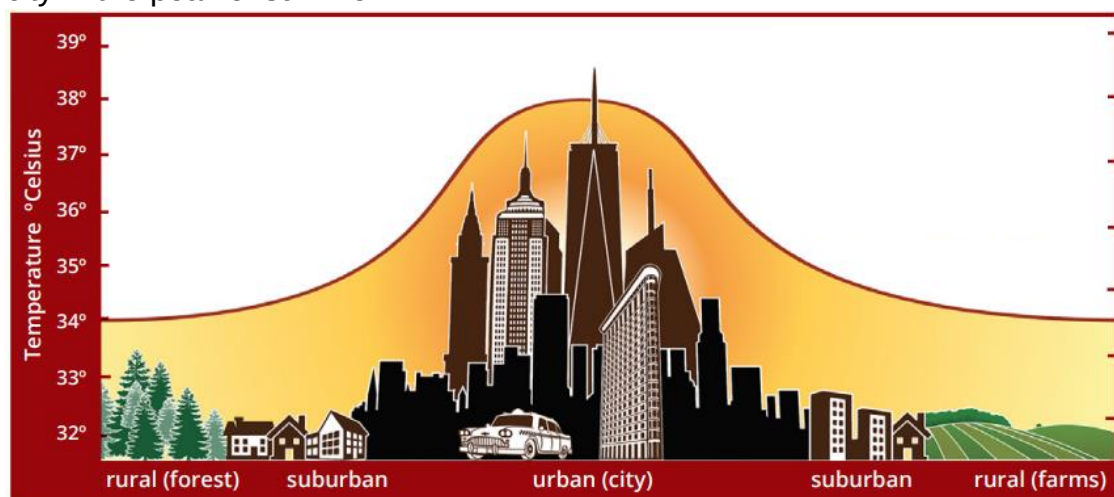


tropical depression, with a localized center close to the border between Mozambique and the southern tip of Malawi, a landlocked country. Although the system has dissipated, it still generated intense rainfall in the interior of Mozambique and southern Malawi.

Source: United Nations Office for the Coordination of Humanitarian Affairs (UNOCHA)

- 1.3.1 What is another term for a tropical cyclone? (1 x 1) (1)
- 1.3.2 What number cyclone was tropical cyclone Freddy in this cyclone season? (1 x 2) (2)
- 1.3.3 “Freddy moved over land as a tropical depression...”. What is a tropical depression? (1 x 2) (2)
- 1.3.4 By referencing the path that Freddy followed, explain what was highly unusual about this tropical cyclone. (2 x 2) (4)
- 1.3.5 List TWO environmental impacts that tropical cyclones can have on a country. (2 x 1) (2)
- 1.3.6 Describe TWO precautions that Mozambique could implement to mitigate the impact of future cyclones. (2 x 2) (4)
- [15]**

- 1.4 Refer to FIGURE 1.4 below showing an urban heat island over a South African city in the peak of summer.



- 1.4.1 What is an *urban heat island*? (1 x 2) (2)
- 1.4.2 Which area will be warmer at night, the urban or rural areas? (1 x 1) (1)
- 1.4.3 “During summer urban areas receive more convectional rain than rural areas”. Defend this statement with an argument explaining why it is correct. (2 x 2) (4)
- 1.4.4 Write a paragraph of approximately EIGHT lines discussing TWO economic consequences of urban heat islands and TWO solutions we can implement to reduce its effect. (4 x 2) (8)

[15]

[40]

QUESTION 2: GEOMORPHOLOGY

2.1. Select the correct answer found in the brackets. Write only the **correct question number** and **answer**. i.e. 2.1.8) Rainfall.

- 2.1.1. Ridges of high ground between individual channels of the stream network are called (interfluvies/confluence points)
- 2.1.2. Rainwater infiltrates and flows through the soil as (throughflow/permeable flow).
- 2.1.3. A drainage pattern that develops on a dome is a (trellis/radial) pattern.
- 2.1.4. The transverse profile is also known as the (longitudinal/cross-profile) of a river.
- 2.1.5. (Isostatic uplift/Abstraction) is the process whereby the watershed becomes lower and its position shifts.

(5 x 1) (5)

2.2.1. State whether the following statements are **TRUE** or **FALSE**. Write only TRUE or FALSE on your answer booklet next to the corresponding number.

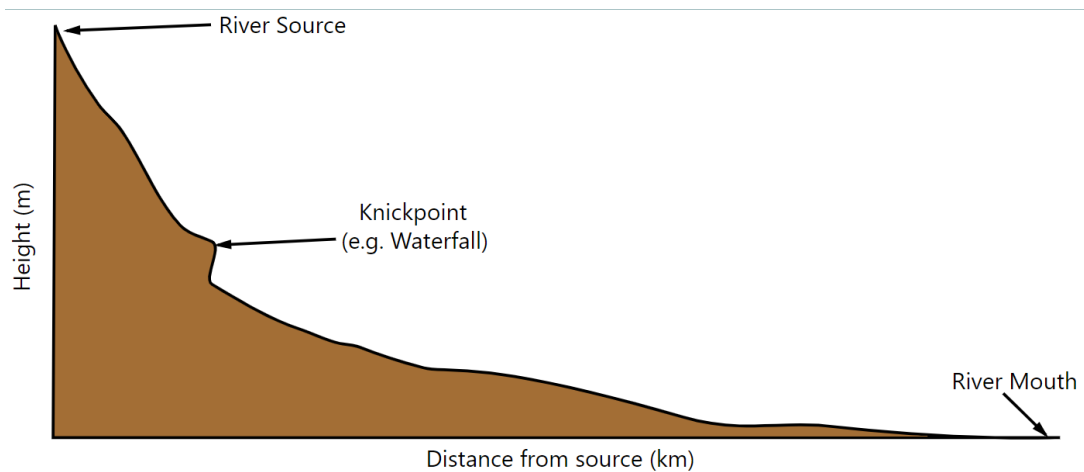
- 2.2.1. Episodic rivers are never supplied with underground water.
- 2.2.2. The correct formula for drainage density is:

$$\frac{\text{Total area of the drainage basin (km}^2\text{)}}{\text{Total length of the streams in the drainage basin (km)}}$$

- 2.2.3. A 4th order stream has a greater volume than a 1st order stream.
- 2.2.4. Rapids are a feature of the upper course of a river.
- 2.2.5. A graded river is convex in shape due to a steep gradient in the upper course and a gentle gradient in the lower course.

(5 x 1) (5)

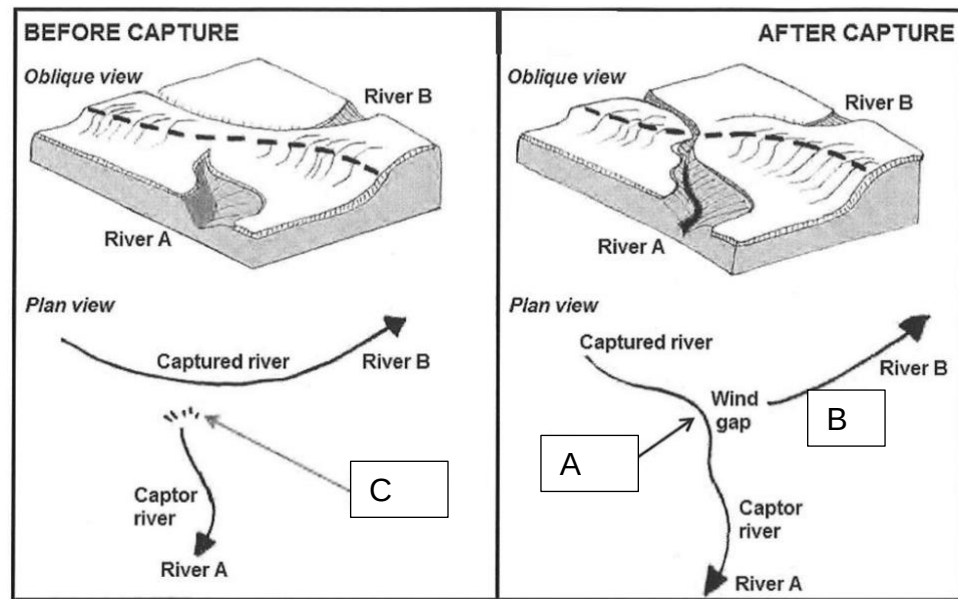
2.3 Refer to FIGURE 2.3. showing a river profile



- 2.3.1 Define the term *longitudinal profile*. (1 x 2) (2)
- 2.3.2 Redraw FIGURE 2.3 above showing the upper, middle and lower courses respectively. NOTE: You will be marked on approximate placement of these labels, they do not have to be exact. (3 x 1) (3)
- 2.3.3 Would the above profile be an example of an *overgraded* or *undergraded* river? (1 x 1) (1)
- 2.3.4 Provide ONE example of a temporary base level from FIGURE 2.3 above. (1 x 1) (1)
- 2.3.5 Differentiate between *turbulent* and *laminar flow*. (2 x 2) (4)
- 2.3.6 Draw a fully labelled **cross section** of a meandering river. Your diagram should include the following labels:
 Slowest current
 Fastest current
 Slip-off slope
 Undercutting (4 x 1) (4)

[15]

2.4 Refer to FIGURE 2.4 showing river capture below.



2.4.1 Define the term *rejuvenation*. (1 x 2) (2)

2.4.2 Describe the process happening at C. (1 x 2) (2)

2.4.3 Write a short paragraph analysing the impact of river capture on both human settlement and the natural wilderness living downstream of River B. (4 x 2) (8)

After capture, River A will now have more energy and be able to erode through more resistant rock. As time wears on this more power river maintains its course, cutting through resistant rock, while the landscape around it changes. Due to its power this river keeps its shape. River A is now at odds with the geology and relief around it.

2.4.4 Is the above description that of a *superimposed drainage pattern* or an *antecedent drainage pattern*? (1 x 1) (1)

2.4.5 Is it possible to identify whether a drainage pattern is superimposed or antecedent without some prior knowledge of the geological history of the area? Justify your answer. (1 + 1) (2)

[15]

[40]

QUESTION 3: SETTLEMENT GEOGRAPHY

3.1. Various possible answers are provided for each question. Write down **only the letter** of the correct answer next to the corresponding number.

3.1.1. In rural areas industrialisation and the modernisation of agricultural practices has led to:

- A. Voluntary resettlement
- B. Less output
- C. Rural-urban migration
- D. Basic need philosophy
- E. Formal resettlement (1)

3.1.2. Which of the following factors is NOT a site factor to consider in the development of a settlement

- A. Availability of water
- B. Rich soils
- C. Flat gradients
- D. Central point of trade
- E. A natural harbour (1)

3.1.3. Which is the smallest settlement?

- A. Hamlet
- B. Village
- C. Isolated farmstead
- D. Conurbation
- E. City (1)

3.1.4. Which of the following would be considered a high-order good?

- A. An imported sports car
- B. A cheap cell phone
- C. Bread
- D. A delicious pizza
- E. A pair of shoes (1)

3.1.5. Rural economic development is hindered because of:

- A. These areas are often remote
- B. Transport costs are expensive
- C. These areas are seen as poor investments
- D. Many workers suffer from diseases
- E. All of the above

(1)

3.2. State whether the following statements are **TRUE** or **FALSE**. Write only TRUE or FALSE next to the corresponding number in your answer book.

3.2.1. Urban areas are multifunctional in nature.

3.2.2. The site of a settlement means a settlement's location in relation to its surrounding features.

3.2.3. Rapid urbanization is most prevalent in more economically developed countries.

3.2.4. Large scale solar farms would best be built in an urban area.

3.2.5. One of the causes of rural-urban migration is a decrease in yield per unit area of agricultural land.

(5x1)(5)

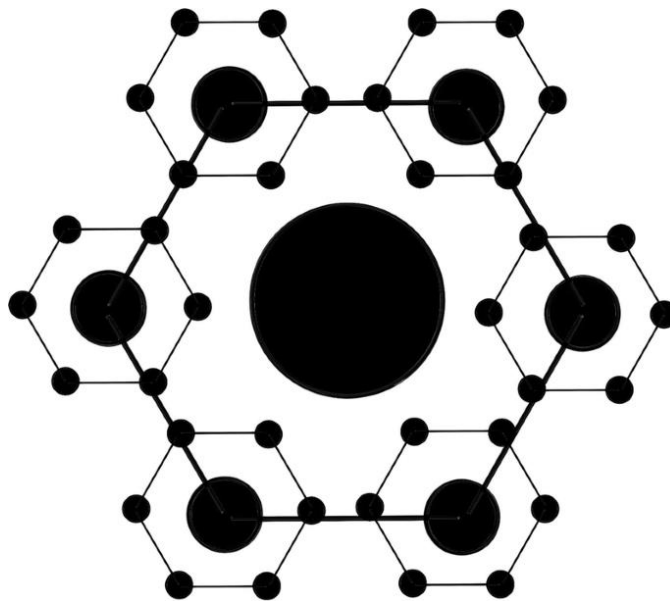
3.3 Refer to FIGURE 3.3 below and answer the questions that follow.



- 3.3.1 Is FIGURE 3.3 referring to rural or urban settlements? (1 x 1) (1)
- 3.3.2 Explain the humour in the cartoon. (1 x 2) (2)
- 3.3.3 The conditions visible in this cartoon have negatively impacted on many site factors that initially contributed to the development of the rural settlement here. List TWO site factors that are being negatively impacted by the conditions seen in the cartoon. (2 x 1) (2)
- 3.3.4 What is *social justice*? (1 x 2) (2)
- 3.3.5 In a paragraph discuss TWO social injustices that many rural South Africans are facing and TWO possible reasons for this. (4 x 2) (8)
- [15]**

3.4 Study FIGURE 3.4 below and answer the questions that follow.

Central Place Theory



- 3.4.1 What is *Central Place Theory*? (1 x 2) (2)
- 3.4.2 Using the image as a guide, draw a legend/key for this diagram indicating the following labels:
Large City
Suburb
Intermediate Centre (3 x 1) (3)
- 3.4.3 Central Place Theory operates under the premise of low-order and high-order centres. Which type of centre would you classify Johannesburg as? (1 x 1) (1)

- 3.4.4 Provide a reason for your answer to QUESTION 3.4.3. (1 x 2) (2)
- 3.4.5 Differentiate between *threshold population* and *sphere of influence*. (2 x 2) (4)
- 3.4.6 You have inherited a lot of money and wish to open an exclusive diamond jewellery store. An estate agent has presented you two locations namely Sandton and Hillbrow. Which location would you choose to set up your store? (1 x 1) (1)
- 3.4.7 List TWO centripetal forces that would have influenced your decision to QUESTION 3.4.6 (2 x 1) (2)
- [15]**

TOTAL SECTION A: 120 MARKS

SECTION B: GEOGRAPHICAL SKILLS AND TECHNIQUES (MAPWORK)

RESOURCE MATERIAL

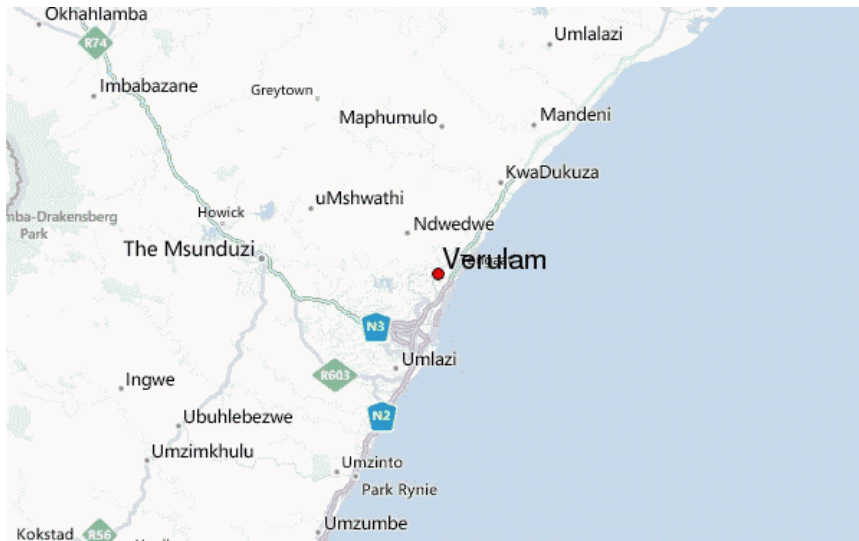
1. An extract from the topographic map 2931 CA VERULAM
2. Orthophoto map 2931 CA 11 VERULAM

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions in the spaces provided in this question paper.
2. Show ALL calculations and formulae, where applicable. Marks will be allocated for these.
3. Indicate the **unit of measurement or compass direction** in the final answer of calculations, e.g. 10 km; 2,1 cm; west of true north.
4. You may use a non-programmable calculator.
5. The area demarcated in RED on the topographic map represents the area covered by the orthophoto map.
6. The following English terms and their Afrikaans translations are shown on the topographic map:

ENGLISH	AFRIKAANS
Diggings	Uitgrawings
River	Rivier
Cemetery	Begraafplaas
Protected Natural Environment	Beskermde Natuurlike Omgewing
Technical College	Tegniese Kollege

GENERAL INFORMATION ON VERULAM



Nestled in the rolling hills of the KwaZulu-Natal coastline, Verulam lies on the banks of the uMdloti River, with the most of the town lying south of the river except the industrial suburbs of Canelands and Barrs Flats. The town is flanked by Waterloo and Ottawa South to the south-east, Phoenix to the south, Mawothi to the south-west and Redcliffe to the west. Neighbouring communities in the greater surrounding area include Mount Edgecombe (10 km) to the south-east, eMdloti (11 km) to the east, uMhlanga (12 km) to the south-east and oThongathi (13km) to the north-east.

Verulam is one of the smaller industrial nodes of the Greater Durban metropolitan area with many industries scattered around the town, mostly concentrated to the north of the town. Notable industries that operate from Verulam include: Shoprite (regional distribution centre), Freshmark- a subsidiary of Shoprite (regional distribution centre), Grafton Everest (furniture), Colgate-Palmolive (oral care), AfriSam (construction materials), Frimax (snacks), Packo (spices) and the Parachute Industries of Southern Africa (sport parachutes).

4.1 MAP SKILLS AND CALCULATIONS

- 4.1.1. Calculate the area covered by the orthophoto map in meters as shown on the topographic map (red block).

Formula: AREA = LENGTH X BREADTH

(4x1) (4)

Refer to the orthophoto map and answer the following questions on gradient.

- 4.1.2. Determine the average gradient of the slope for a truck that will transport its cargo (goods) from Spot Height 29 at the industries near 6 (block A5) to Spot Height 164 in Everest heights in block E1 on the **orthophoto** map. Show ALL calculations.

Formula: Average gradient = $\frac{\text{Vertical interval (VI)}}{\text{Horizontal equivalent (HE)}}$

OR

Formula: Difference in Height : Horizontal Distance

(4x1) (4)

- 4.1.3. Will it be easy or difficult for the truck to transport its cargo over the calculated gradient? (1x2) (2)

[10]

4.2 MAP INTERPRETATION

- 4.2.1 Refer to block **D5** on the topographic map. What direction is the Mdloti river flowing? (1 x 1) (1)
- 4.2.2 Provide a reason evident on the map to justify your answer to QUESTION 4.2.1. (1 x 2) (2)
- 4.2.3 Refer to block **E4**. What drainage basin pattern is evident for this river system? (1 x 1) (1)
- 4.2.4 Refer to block **E4**. Give the stream order of the river exiting block E4. (1 x 2) (2)
- 4.2.5 Refer to the settlement of **Hazelmere (A2)** and **Verulam (South west corner of the map)**.
- (a) Which of these settlements would be classified as a town and which settlement would be viewed as a village? (2 x 1) (2)
 - (b) Which settlement would have a larger sphere of influence? (1 x 1) (1)
 - (c) Which settlement would have the correct threshold population to open a Law Firm? (1 x 1) (1)

- 4.2.6 **K** indicates an area of Verulam that has expanded rapidly in the last few years. Discuss ONE issue these residents may face related to rapid urbanisation. (1 x 2) (2)
[12]

4.3 GEOGRAPHICAL INFORMATION SYSTEMS (GIS)

- 4.3.1 (a) Define the term *Buffering*. (1 x 2) (2)
- (b) Give evidence to show that buffering was implemented in blocks **B5 and C5** around the airport. (1 x 1) (1)
- 4.4.3 The Orthophoto map 2931 CA 11 VERULAM is secondary data.
- (a) What is *secondary data*? (1 x 2) (2)
- (b) Give one advantage of using secondary data. (1 x 2) (2)
- (c) Is the orthophoto map an example of raster or vector data? (1 x 1) (1)
[8]

TOTAL SECTION B: 30 MARKS

Grand Total: 150 Marks